CFA® 2026 Level I

Complete Study Guide

Financial Statement Analysis & Equity

This document is a condensed study guide. Please refer to the official CFA Institute curriculum for complete details.

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Part I

Financial Statement Analysis

1 Introduction to Financial Statement Analysis

1.1 Financial Statement Analysis Framework

LOS 27.a: The Six-Step Framework:

Step 1: State Objective and Context. Determine what questions the analysis seeks to answer, the form of presentation needed, and available resources and time.

Step 2: Gather Data. Acquire financial statements and relevant industry and economic data. Interview management, suppliers, and customers.

Step 3: Process Data. Make adjustments to financial statements. Calculate ratios and perform statistical analysis. Prepare graphs and common-size statements.

Step 4: Analyze and Interpret Data. Use processed data to answer the questions from Step 1. Determine supported conclusions or recommendations.

Step 5: Report Conclusions. Prepare and communicate report in compliance with Code and Standards.

Step 6: Update Analysis. Repeat steps periodically, changing conclusions as necessary.

LOS 27.b: Roles of Financial Statement Analysis. Financial reporting shows performance to investors and creditors. Financial statement analysis uses this information to make economic decisions: whether to invest in securities, extend credit, or assign credit ratings. Analysts evaluate past performance and current position to form opinions about future profitability and cash flow.

Note: *Management also performs financial analysis for internal decisions, but uses internal information unavailable to external users.*

LOS 27.c: Standard-Setting and Regulatory Bodies.

Standard-setting bodies establish financial reporting standards:

- **FASB:** U.S. Generally Accepted Accounting Principles (U.S. GAAP)
- **IASB:** International Financial Reporting Standards (IFRS)

Regulatory authorities enforce compliance:

- **SEC:** U.S. Securities and Exchange Commission
- **FCA:** Financial Conduct Authority (UK)
- **IOSCO:** International Organization of Securities Commissions (coordinates global regulation)

The Sarbanes-Oxley Act (2002) prohibits auditors from providing certain paid services, requires executive certification of financial statements, and requires internal controls effectiveness statements.

Key SEC Filings:

Form	Description
S-1	Registration statement for new securities; includes audited financial statements
10-K	Annual filing with audited financial statements, business risks, management discussion
10-Q	Quarterly filing with unaudited interim financial statements
DEF-14A	Proxy statement for shareholder votes; includes board elections, compensation
8-K	Material events disclosure (acquisitions, management changes, accounting matters)
3,4,5	Beneficial ownership of securities by officers and directors

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Financial Statement Notes (Footnotes):

- Discuss basis of presentation (fiscal period, GAAP/IFRS, consolidation)
- Provide accounting methods, assumptions, and estimates
- Include acquisitions, legal actions, employee benefits, contingencies, related parties, segments
- Are audited along with primary statements

A **business segment** accounts for ¿10% of revenues, assets, or income, and has distinguishable risk/return characteristics. Reported segments must account for at least 75% of external sales. Required disclosures: revenue (external and between segments), profit/loss, assets/liabilities, interest, PP&E acquisitions, depreciation, other noncash expenses, income tax.

Management Commentary (MD&A): Required by SEC to discuss trends affecting liquidity, capital resources, and operations; effects of inflation; off-balance-sheet obligations; accounting policies requiring judgment; forward-looking expenditures.

Audit Opinions:

- **Unqualified (unmodified):** Clean opinion; statements free from material errors
- **Qualified:** Exceptions to accounting principles noted
- **Adverse:** Statements not fairly presented
- **Disclaimer:** Auditor unable to express opinion

Auditor's opinion includes an explanatory paragraph when material loss is probable but amount cannot be estimated (going concern issues, asset valuation, litigation).

LOS 27.d: IFRS vs U.S. GAAP Differences:

Item	U.S. GAAP	IFRS
Philosophy	Rules-based	Principles-based
Inventory	LIFO allowed	LIFO prohibited
Development costs	Expensed	May capitalize
Interest paid	CFO	CFO or CFF
Inventory write-down reversal	Prohibited	Allowed

LOS 27.e: Information Sources:

- **Issuer:** Earnings calls, press releases, management communications
- **Public third-party:** Industry reports, government statistics, media, social media
- **Proprietary third-party:** Analyst reports, Bloomberg, FactSet
- **Proprietary primary:** Commissioned studies, hands-on experience, technical specialists

2 Analyzing Income Statements

2.1 Revenue Recognition

LOS 28.a: Revenue Recognition Principles. Revenue is recognized when earned, not necessarily when cash is collected. Revenue is reported net of returns and allowances. If payment is received before transfer, unearned revenue (liability) is created.

Five-Step Process (IFRS 15/ASC 606):

1. Identify the contract with a customer
2. Identify separate performance obligations
3. Determine the transaction price
4. Allocate price to performance obligations
5. Recognize revenue when performance obligation is satisfied

A **performance obligation** is a promise to deliver a distinct good or service. A distinct good/service meets two criteria: (1) customer can benefit from it on its own or with readily available resources, and (2) the promise is separately identifiable.

A performance obligation is satisfied over time if any of three criteria are met: (1) customer receives benefits as supplier performs, (2) supplier creates asset customer controls, or (3) asset has no alternative use and supplier has right to payment for work completed.

Example: Long-term contract: Builder agrees to \$10M warehouse with \$8M estimated costs. Year 1 costs \$4M (50% complete). Revenue recognized = \$5M (50% of \$10M).

Example: Agent vs Principal: Travel agent sells \$10,000 ticket, receives \$1,000 commission. As agent, revenue = \$1,000 (net). As principal, revenue = \$10,000 with \$9,000 expense.

Example: Bill-and-hold: Revenue may be recognized before shipping if: customer requested arrangement, goods are identified as belonging to customer, goods are complete and ready, goods cannot be redirected.

2.2 Expense Recognition

LOS 28.b: Expense Recognition Methods:

Matching Principle: Expenses to generate revenue are recognized in the same period as the revenue. Examples: COGS when inventory sold, warranty expense at time of sale.

Capitalization: Costs initially recorded as assets, then expensed through depreciation or amortization over the asset's life. Applies to PP&E and finite-lived intangibles.

Period Costs: Costs not directly tied to revenue generation (e.g., administrative costs) expensed when incurred.

Capitalizing vs Expensing Effects:

Impact	Capitalizing	Expensing
Assets & Equity	Higher	Lower
Net income (Year 1)	Higher	Lower
Net income (subsequent years)	Lower	Higher
ROA & ROE (Year 1)	Higher	Lower
Debt-to-equity	Lower	Higher
CFO	Higher	Lower
CFI	Lower	Higher

Example: Capitalizing vs expensing: Equipment \$250,000 + \$10,000 installation = \$260,000 capitalized. Training \$7,500 + repairs \$35,000 = \$42,500 expensed.

Capitalized Interest: Interest during construction of an asset for own use is capitalized, not expensed. It is allocated through depreciation. For analysis, both capitalized and expensed interest should be used in coverage ratios.

Example: Interest coverage adjustment: EBIT €160M, interest €80M, capitalized interest €20M, depreciation of capitalized interest €10M. Adjusted ratio = (160+10)/(80+20) = 1.7 (down from 2.0).

Research & Development:

- **IFRS:** Research expensed; development may be capitalized if criteria met (technical feasibility, intention to complete, market exists, resources available)
- **U.S. GAAP:** R&D generally expensed; software development capitalized after technological feasibility

2.3 Nonrecurring Items

LOS 28.c: Nonrecurring Items Classification:

Unusual or Infrequent Items: Gains/losses from asset sales, impairments, restructuring costs. Reported before tax in continuing operations.

Discontinued Operations: Business must be physically and operationally distinct. Reported separately, net of tax, after continuing operations. Past statements restated.

Accounting Changes:

- **Policy change:** Retrospective application (restate prior periods)
- **Estimate change:** Prospective application (no restatement)
- **Error correction:** Prior-period adjustment (retrospective)

2.4 Earnings Per Share

LOS 28.d: Earnings Per Share (EPS).

$$\text{Basic EPS} = \text{Net Income - Preferred Dividends}_{\overline{\text{Weighted Average Common Shares Outstanding}}}$$

Stock Dividends and Splits: Applied retroactively to beginning of year. Each shareholder's proportional ownership unchanged.

Example: Weighted average shares: Beginning 10,000 shares; April 1 issue 4,000; July 1 10% stock dividend; Sept 1 repurchase 3,000.

- Jan 1: 10,000 × 1.10 × 12/12 = 11,000
- Apr 1: 4,000 × 1.10 × 9/12 = 3,300
- Sep 1: -3,000 × 4/12 = -1,000
- Total: 13,300 shares

$$\text{Diluted EPS} = \text{NI - Pfd Div + Convertible Pfd Div +}$$
$$\text{After-tax Interest}_{\overline{\text{WA Shares+Conversion Shares+Option Shares}}}$$

Example: Convertible preferred: NI \$4.35M, 2M shares, \$5M 7% preferred convertible to 550,000 shares. Basic EPS = (\$4.35M - \$0.35M)/2M = \$2.00. Diluted EPS = \$4.35M/(2M + 0.55M) = \$1.71.

Example: Convertible debt: Earnings \$2.5M, 1M shares, 2,000 bonds convertible to 120 shares each, 5% interest, 30% tax. Shares = 240,000. Add back interest = \$70,000. Diluted EPS = (\$2.5M + \$0.07M)/(1M + 0.24M) = \$2.07.

Treasury Stock Method for Options/Warrants:

$$\text{Net Increase in Shares} = \left(\frac{\text{AMP}-\text{EP}}{\text{AMP}}\right) \times N$$

Example: Stock options: 100,000 options at \$15, avg market price \$20. Net increase = (20-15)/20 × 100,000 = 25,000 shares. Diluted EPS = \$1.2M/(500,000 + 25,000) = \$2.29.

2.5 Ratios and Common-Size Statements

LOS 28.e: Common-Size Income Statement: All items as percentage of revenue.

$$\boxed{\text{Gross Margin} = \text{Gross Profit}_{\overline{\text{Revenue}}}}$$

$$\boxed{\text{Operating Margin} = \text{EBIT}_{\overline{\text{Revenue}}}}$$

$$\boxed{\text{Net Margin} = \text{Net Income}_{\overline{\text{Revenue}}}}$$

$$\boxed{\text{Pretax Margin} = \text{EBT}_{\overline{\text{Revenue}}}}$$

3 Analyzing Balance Sheets

3.1 Intangible Assets and Goodwill

LOS 29.a: Intangible Assets. Identifiable: patents, trademarks, copyrights (amortized). **Unidentifiable:** goodwill (not amortized, impairment tested). IFRS: research expensed, development may be capitalized. U.S. GAAP: R&D generally expensed.

LOS 29.b: Goodwill.

$$\text{Goodwill} = \text{Purchase Price - Fair Value of Identifiable Net Assets}$$

Goodwill is not amortized; tested for impairment annually. Impairment loss recognized in income statement.

3.2 Financial Instruments

LOS 29.c: Financial Instrument Classification:

Type	**U.S. GAAP**	**IFRS**
Held-to-maturity	Amortized cost	Amortized cost
Trading	Fair value (P&L)	Fair value (P&L)
Available-for-sale	Fair value (OCI)	Fair value (OCI)

3.3 Common-Size Balance Sheet

LOS 29.e: Common-Size Balance Sheet: All items as percentage of total assets.

Liquidity Ratios:

$$\text{Current Ratio} = \text{CA}_{\overline{\text{CL}}}$$

$$\text{Quick Ratio} = \text{Cash} + \text{MS} + \text{AR}_{\overline{\text{CL}}}$$

$$\text{Cash Ratio} = \text{Cash} + \text{MS}_{\overline{\text{CL}}}$$

Solvency Ratios:

$$\text{Debt-to-Equity} = \text{Total Debt}_{\overline{\text{Total Equity}}}$$

$$\text{Debt-to-Assets} = \text{Total Debt}_{\overline{\text{Total Assets}}}$$

$$\text{Financial Leverage} = \text{Total Assets}_{\overline{\text{Total Equity}}}$$

4 Analyzing Statements of Cash Flows I

4.1 Cash Flow Statement Basics

LOS 30.a: Cash Flow Statement Links.

$$\text{CFO} + \text{CFI} + \text{CFF} = \Delta\text{Cash}$$

Quality of earnings is high when CFO NI. If NI consistently exceeds CFO, earnings quality is lower.

4.2 Direct and Indirect Methods

LOS 30.b: Direct Method: Shows actual cash receipts and payments. Components: cash from customers, cash paid to suppliers, cash operating expenses, interest paid, taxes paid.

Indirect Method:

$$\text{CFO} = \text{NI} + \text{NCC} - \text{WC}_{\text{INV}}$$

Noncash charges: depreciation, amortization, losses, increase in DTL. Subtract: gains, increase in DTA.

Working capital adjustments: subtract increases in operating assets, add decreases; add increases in operating liabilities, subtract decreases.

Example: CFO calculation: NI \$39,000 + Depreciation \$7,000 + Increase DTL \$5,000 + Loss \$2,000 - Gain \$10,000 - Increase AR \$1,000 + Decrease Inv \$2,000 + Increase AP \$4,000 - Decrease WP \$3,500 + Increase Interest Payable \$500 + Increase Unearned Revenue \$4,000 + Increase TP \$1,000 = \$50,000.

4.3 Investing and Financing Activities

CFI: Purchase/sale of PP&E, intangibles, debt/equity investments, loans made to others.

CFF: Debt issuance/repayment, stock issuance/repurchase, dividends paid.

Example: Computing CFI: PP&E purchased \$25,000. Disposal: cost \$16,000, accumulated deprec. \$4,000, loss \$2,000 → proceeds \$10,000. Land disposal: carrying value \$5,000, gain \$10,000 → proceeds \$15,000. CFI = -\$25,000 + \$10,000 + \$15,000 = \$0.

Example: Computing CFF: Bonds payable +\$5,000. Contributed capital -\$10,000 (share repurchase). Dividends declared \$8,500, increase in dividends payable \$5,000 → cash dividends \$3,500. CFF = \$5,000 - \$10,000 - \$3,500 = -\$8,500.

LOS 30.d: IFRS vs U.S. GAAP:

Item	U.S. GAAP	IFRS
Interest received	CFO	CFO or CFI
Interest paid	CFO	CFO or CFF
Dividends received	CFO	CFO or CFI
Dividends paid	CFF	CFO or CFF
Taxes	CFO	May be split

5 Analyzing Statements of Cash Flows II

LOS 31.b: Free Cash Flow.

$$\text{FCFF} = \text{CFO} + [\text{Int} \times (1 - t)] - \text{FC}_{\text{INV}}$$

$$\text{FCFE} = \text{CFO - FC}_{\text{INV}} + \text{Net Borrowing}$$

Cash Flow Ratios:

$$\text{Cash Flow-to-Revenue} = \text{CFO}_{\overline{\text{Revenue}}}$$

$$\text{Cash ROA} = \text{CFO}_{\overline{\text{Avg Assets}}}$$

$$\text{Cash ROE} = \text{CFO}_{\overline{\text{Avg Equity}}}$$

$$\text{Debt Coverage} = \text{CFO}_{\overline{\text{Total Debt}}}$$

$$\text{Interest Coverage} = \text{CFO} + \text{Interest Paid} + \text{Taxes Paid}_{\overline{\text{Interest Paid}}}$$

6 Analysis of Inventories

LOS 32.a: Inventory Valuation:

- **IFRS:** Lower of cost or NRV (write-ups allowed)
- **U.S. GAAP:** Lower of cost or market for LIFO users; no write-ups

A write-down decreases inventory, assets, equity; increases turnover and debt ratios; decreases margins, ROA, ROE.

LOS 32.b: FIFO vs LIFO (Inflation):

	FIFO	**LIFO**
COGS	Lower	Higher
Ending Inventory	Higher	Lower
Gross Profit	Higher	Lower
Current Ratio	Higher	Lower
Inventory Turnover	Lower	Higher
Debt-to-Equity	Lower	Higher

LIFO Liquidation: When inventory declines, older lower costs flow to COGS, inflating profits (unsustainable).

LOS 32.c: Inventory Ratios:

$$\text{Inventory Turnover} = \text{COGS}_{\overline{\text{Avg Inventory}}}$$

$$\text{DOH} = 365_{\overline{\text{Inventory Turnover}}}$$

Increasing finished goods while raw materials decrease may indicate decreasing demand.

7 Analysis of Long-Term Assets

LOS 33.a: Intangible Asset Reporting:

- Purchased: Capitalized
- Internally developed: Generally expensed (IFRS allows development capitalization)
- Acquired in business combination: Fair value; excess is goodwill

LOS 33.b: Impairment:

- **IFRS:** Carrying value ¿ Recoverable amount (max of FV - selling costs and value in use). Write down; reversals allowed.
- **U.S. GAAP:** Two-step: (1) Carrying value ¿ undiscounted CF; (2) Loss = carrying value - fair value. No reversals.

Example: Impairment: Carrying value $800,000; undiscounted CF $795,000; fair value $790,000; value in use $785,000. U.S. GAAP: impaired, loss $10,000. IFRS: recoverable amount $785,000, loss $15,000.

LOS 33.c: Asset Age Calculations:

$$\boxed{\text{Average Age} = \text{Accumulated Depreciation}\frac{}{\text{Annual Depreciation}}}$$

$$\boxed{\text{Remaining Life} = \text{Net PP\&E}\frac{}{\text{Annual Depreciation}}}$$

$$\boxed{\text{Total Life} = \text{Gross PP\&E}\frac{}{\text{Annual Depreciation}}}$$

8 Topics in Long-Term Liabilities and Equity

LOS 34.a: Lease Classification (Finance if ANY met):

1. Ownership transfers
2. Bargain purchase option
3. Lease term 75% of useful life
4. PV of payments 90% of fair value
5. Asset is specialized

Lessee Accounting: Record ROU asset and lease liability at PV of payments. Interest and amortization expense (finance lease). Operating lease (U.S. GAAP): single lease expense.

LOS 34.b: Pension Plans:

- **Defined Contribution:** Expense = contribution; no balance sheet liability
- **Defined Benefit:** Funded status (plan assets - PBO) on balance sheet

Stock-Based Compensation: Expense fair value at grant date over vesting period. Stock options use option pricing models.

9 Analysis of Income Taxes

LOS 35.a: Deferred Tax Definitions:

- **DTL:** Tax expense ¿ Taxes payable (future cash outflow)
- **DTA:** Taxes payable ¿ Tax expense (future tax savings)
- **Permanent difference:** Never reverses; no DTA/DTL

$$\boxed{\text{Tax Expense} = \text{Taxes Payable} + \Delta\text{DTL} - \Delta\text{DTA}}$$

LOS 35.c: Tax Rates:

$$\boxed{\text{Effective Rate} = \text{Tax Expense}_{\overline{\text{Pretax Income}}}}$$

$$\boxed{\text{Cash Rate} = \text{Cash Taxes Paid}_{\overline{\text{Pretax Income}}}}$$

10 Financial Reporting Quality

LOS 36.a-c: Quality Spectrum:

1. GAAP compliant, decision useful, sustainable earnings
2. GAAP compliant, decision useful, low earnings quality
3. GAAP compliant, biased choices
4. Earnings actively managed
5. Not GAAP compliant but based on actual activities
6. Fraudulent

Aggressive vs Conservative:

Aggressive	**Conservative**
Capitalize costs	Expense costs
Longer asset lives	Shorter asset lives
Higher salvage value	Lower salvage value
Straight-line	Accelerated depreciation
Lower bad debt reserve	Higher bad debt reserve

LOS 36.g: Earnings Management Methods:

- Revenue recognition (channel stuffing, bill-and-hold)
- Estimates (bad debt, warranty, valuation allowance)
- Depreciation (methods, useful lives)
- Inventory cost flow (FIFO/LIFO)
- Capitalization decisions
- Related-party transactions

LOS 36.h: Warning Signs:

- Revenue growth out of line with peers
- Decreasing turnover ratios
- CFO persistently ¡ NI
- Unusual capitalization
- Frequent nonrecurring items
- Emphasis on non-GAAP measures

11 Financial Analysis Techniques

11.1 Financial Ratios

LOS 37.b: Activity Ratios:

$$\text{Receivables Turnover} = \text{Revenue}_{\overline{\text{Avg Receivables}}}$$

$$\text{DSO} = 365_{\overline{\text{Receivables Turnover}}}$$

$$\text{Inventory Turnover} = \text{COGS}_{\overline{\text{Avg Inventory}}}$$

$$\text{DOH} = 365_{\overline{\text{Inventory Turnover}}}$$

$$\text{Payables Turnover} = \text{Purchases}_{\overline{\text{Avg AP}}}$$

$$\text{DPO} = 365_{\overline{\text{Payables Turnover}}}$$

$$\text{Total Asset Turnover} = \text{Revenue}_{\overline{\text{Avg Assets}}}$$

$$\text{Cash Conversion Cycle} = \text{DOH} + \text{DSO - DPO}$$

Liquidity Ratios:

$$\text{Current Ratio} = \text{CA}_{\overline{\text{CL}}}$$

$$\text{Quick Ratio} = \text{Cash} + \text{MS} + \text{AR}_{\overline{\text{CL}}}$$

Solvency Ratios:

$$\text{Debt-to-Equity} = \text{Total Debt}_{\overline{\text{Total Equity}}}$$

$$\text{Debt-to-Assets} = \text{Total Debt}_{\overline{\text{Total Assets}}}$$

$$\text{Interest Coverage} = \text{EBIT}_{\overline{\text{Interest}}}$$

$$\text{Fixed Charge Coverage} = \text{EBIT} + \text{Lease}_{\overline{\text{Interest+Lease}}}$$

Profitability Ratios:

$$\text{Gross Margin} = \text{Gross Profit}_{\overline{\text{Revenue}}}$$

$$\text{Operating Margin} = \text{EBIT}_{\overline{\text{Revenue}}}$$

$$\text{Net Margin} = \text{Net Income}_{\overline{\text{Revenue}}}$$

$$\text{ROA} = \text{Net Income}_{\overline{\text{Avg Assets}}}$$

$$\text{ROE} = \text{Net Income}_{\overline{\text{Avg Equity}}}$$

11.2 DuPont Analysis

LOS 37.d: DuPont Decomposition:

$$\text{ROE} = \text{Net Margin} \times \text{Asset Turnover} \times \text{Leverage}$$

$$\text{ROE} = \text{NI}_{\overline{\text{Rev} \times \frac{\text{Rev}}{\text{Assets}} \times \frac{\text{Assets}}{\text{Equity}}}}$$

Extended DuPont (5-way):

$$\text{ROE} = \text{NI}_{\overline{\text{EBT} \times \frac{\text{EBT}}{\text{EBIT}} \times \frac{\text{EBIT}}{\text{Rev}} \times \frac{\text{Rev}}{\text{Assets}} \times \frac{\text{Assets}}{\text{Equity}}}}$$

12 Introduction to Financial Statement Modeling

LOS 38.a: Pro Forma Model Steps:

1. Estimate revenue growth
2. Estimate COGS (% of sales)
3. Estimate SG&A
4. Estimate financing costs
5. Estimate tax expense
6. Model balance sheet
7. Estimate capital expenditures
8. Construct pro forma cash flow

LOS 38.b: Behavioral Biases:

- Overconfidence: Narrow confidence intervals; share forecasts, evaluate past errors
- Illusion of control: Overcomplicating models; focus on known explanatory variables
- Conservatism (anchoring): Small adjustments to prior forecasts; periodic error evaluation
- Representativeness: Base-rate neglect; consider both inside and outside views
- Confirmation bias: Seek contrary views; consult uninvolved colleagues

LOS 38.c: Porter's Five Forces:

1. Threat of new entrants (barriers to entry)
2. Threat of substitutes (availability)
3. Bargaining power of buyers (concentration, switching costs)
4. Bargaining power of suppliers (scarcity, concentration)
5. Rivalry among competitors (number of firms, fixed costs, exit barriers)

LOS 38.e: Forecast Horizon: For buy-side: expected holding period. For cyclical companies: include a full business cycle (midcycle normalized earnings). For acquisitions/restructurings: long enough to realize benefits.

Part II

Equity Investments

13 Market Organization and Structure

LOS 39.a: Three Main Functions:

1. Allow saving, borrowing, raising equity, risk management, trading, information utilization
2. Determine returns equating savings supply and borrowing demand
3. Allocate capital to most efficient uses

LOS 39.b: Asset Classifications:

- Financial vs Real assets
- Debt vs Equity
- Public vs Private
- Spot vs Futures
- Primary vs Secondary
- Money vs Capital markets
- Traditional vs Alternative

LOS 39.c: Major Asset Types:

- **Securities:** Fixed-income (bonds, notes, commercial paper), Equity (common, preferred, warrants), Pooled (mutual funds, ETFs, hedge funds, ABS)
- **Currencies:** Reserve currencies: USD, EUR, GBP, JPY, CHF
- **Contracts:** Forwards (customized, OTC), Futures (standardized, exchange), Options (calls/puts), Swaps, Insurance, CDS
- **Commodities:** Metals, agricultural, energy
- **Real assets:** Real estate, equipment

LOS 39.d: Financial Intermediaries:

- Brokers: Find counterparties
- Dealers: Trade from inventory; profit from bid-ask spread
- Exchanges: Trading venue, regulation
- Securitizers: Pool assets, sell interests (tranches)
- Banks: Deposit taking, lending
- Insurance: Pool risks
- Arbitrageurs: Exploit price differences
- Clearinghouses: Guarantee completion, reduce counterparty risk

LOS 39.e: Long vs Short:

- Long: Own or right to buy; benefits from price increase
- Short: Borrowed and sold; benefits from price decrease

LOS 39.f: Leverage Calculations:

$$\boxed{\text{Leverage Ratio} = \text{Asset Value}_{\overline{\text{Equity}}}}$$

$$\boxed{\text{Margin Call Price} = \text{P}_0 \times \frac{1-\text{IM}}{1-\text{MM}}}$$

Example: Margin transaction: 1,000 shares at \$100, 40% margin, 4% interest, sell at \$110. Initial equity \$40,050. Return = \$9,500/\$40,050 = 23.72%.

Example: Margin call: Purchase \$40, 50% initial margin, 25% maintenance. Price = \$40 × (1-0.5)/(1-0.25) = \$26.67.

LOS 39.g-h: Order Types:

- Market: Execute immediately at best price
- Limit: Execute at specified price or better
- Stop: Becomes market when stop price reached
- Day: Expires end of day
- GTC: Stays until filled
- IOC: Cancel if not filled immediately

LOS 39.i: Primary vs Secondary: Primary issues new securities. Secondary trades existing securities, providing liquidity and price discovery.

LOS 39.j: Market Structures:

- Quote-driven (dealer): Trade with dealers (OTC)
- Order-driven (exchange): Rules match buyers/sellers
- Brokered: Brokers locate counterparties (illiquid assets)

14 Security Market Indexes

LOS 40.a-b: Index Returns:

$$\text{Price Return} = \text{V}_1 - V_0 \overline{V_0}$$

$$\text{Total Return} = \text{V}_1 - V_0 + \text{Income}_{\overline{V_0}}$$

LOS 40.d: Weighting Methods:

- **Price-weighted:** Sum of prices / divisor. Higher price = more weight. DJIA.
- **Equal-weighted:** Arithmetic average of returns. Requires rebalancing.
- **Market cap-weighted:** Weight = market cap / total market cap. Most common. S&P 500.
- **Float-adjusted:** Uses shares available to public.
- **Fundamental-weighted:** Based on earnings, dividends, cash flow.

Example: Price-weighted: \$10, \$20, \$60 → index = 30. After 2-for-1 split of \$60 stock: new price \$45, new divisor = (10+20+45)/30 = 1.875.

Example: Market cap-weighted: Base market cap \$40M, new \$95M → index = (95/40)×100 = 118.75.

LOS 40.f: Rebalancing & Reconstitution: Rebalancing adjusts weights (important for equal-weighted). Reconstitution adds/deletes securities.

LOS 40.g: Uses of Indexes: Market sentiment, manager benchmark, market return/risk measure, beta measurement, model portfolio.

LOS 40.h: Equity Index Types: Broad market, multi-market, sector, style (growth/value, market cap).

LOS 40.j: Fixed-Income Index Issues: Large universe, dealer markets, infrequent trading, high turnover, difficult to replicate.

LOS 40.k: Alternative Indexes:

- Commodity: Based on futures, not spot; weighting varies
- Real estate: Appraisal-based, repeat sales, REIT-based
- Hedge fund: Voluntary reporting creates survivorship/backfill bias

15 Market Efficiency

LOS 41.a: Informationally Efficient Market: Prices fully, quickly, rationally reflect all available information. Only unexpected information moves prices. Passive investing outperforms active in efficient markets.

LOS 41.b: Market Value vs Intrinsic Value: Market value = current price. Intrinsic value = value rational investor with full knowledge would pay.

LOS 41.c: Efficiency Factors:

- More participants ; more efficient
- More information ; more efficient
- Arbitrage/short selling impediments ; less efficient
- High transaction/information costs ; less efficient

LOS 41.d: Forms of Market Efficiency:

Form	Information	Implication
Weak-form	Past price/volume	Technical analysis ineffective
Semi-strong	All public	Fundamental analysis ineffective
Strong	All public & private	No abnormal returns

LOS 41.f: Market Anomalies:

- Calendar: January effect, weekend effect
- Overreaction: Losers outperform winners
- Momentum: High short-term returns continue
- Size: Small-cap outperformance
- Value: Value stocks outperform growth
- Closed-end fund discounts
- Post-announcement drift
- IPO underpricing (long-term underperformance)

LOS 41.g: Behavioral Finance Biases:

- Loss aversion: Dislike losses more than like equal gains
- Overconfidence: Overestimate abilities
- Herding: Follow others' actions
- Information cascade: Mimic informed traders

16 Overview of Equity Securities

LOS 42.a: Types of Equity:

- Common: Residual claim, voting rights, variable dividends
- Preferred: Fixed dividends, priority, usually no voting rights
- Cumulative preferred: Missed dividends accumulate
- Participating preferred: Extra dividends if profits exceed threshold
- Convertible preferred: Exchangeable for common
- Callable: Firm can repurchase (higher risk)
- Putable: Investor can sell back (lower risk)

LOS 42.b: Voting Rights: Statutory: one vote per share per director. Cumulative: votes can be allocated to one candidate, allowing minority representation.

LOS 42.c: Public vs Private Equity: Private equity: less liquid, negotiated prices, limited disclosure, lower costs, longer-term focus, potentially higher returns. Types: venture capital, LBOs, PIPE.

LOS 42.d: Foreign Investment Methods:

- Direct: Buy foreign securities on foreign exchanges
- ADRs: Trade on U.S. exchanges in USD
- GDRs: Issued outside U.S. and issuer's home country
- GRS: Trade in multiple currencies

LOS 42.g: Market vs Book Value: Book value = assets - liabilities (historical). Market value = price × shares (expectations).

LOS 42.h: ROE and Cost of Equity:

$$\boxed{\text{ROE} = \text{NI - Preferred Div}_{\overline{\text{Avg Book Value of Common Equity}}}}$$

17 Company Analysis: Past and Present

LOS 43.a: Research Report Elements: Front matter (recommendation, target prices), rationales, company description, industry overview, financial analysis, valuation, ESG, risks.

LOS 43.c: Pricing Power: Depends on market structure. Competitive markets (many firms, undifferentiated products, low barriers) → low pricing power. Less competitive markets (differentiation, high barriers, high switching costs) → pricing power.

LOS 43.d: Operating Profitability:

$$\boxed{\text{Operating Profit} = \text{Q(P - VC) - FC}}$$

$$\boxed{\text{DOL} = \%\Delta\text{Operating Profit}_{\overline{\%\Delta\text{Sales}}}}$$

$$\boxed{\text{DFL} = \%\Delta\text{Net Income}_{\overline{\%\Delta\text{Operating Income}}}}$$

Economies of scale: increasing output decreases unit costs. Economies of scope: adding product lines decreases unit costs.

18 Industry and Competitive Analysis

LOS 44.b: Industry Classification: GICS: sector → industry group → industry → subindustry. Top 11 sectors: Energy, Financials, Basic Materials, IT, Industrials, Telecom, Consumer Discretionary, Utilities, Consumer Staples, Real Estate, Health Care.

LOS 44.c: Industry Metrics:

- Industry size = total annual sales
- Growth vs mature industries
- Cyclical vs defensive
- Market share = company revenue / industry size
- HHI = sum of squared market shares (0-10,000); higher = less competition

LOS 44.d: Porter's Five Forces:

1. Rivalry among existing competitors
2. Threat of new entrants
3. Threat of substitutes
4. Bargaining power of buyers
5. Bargaining power of suppliers

PESTLE Analysis: Political, Economic, Social, Technological, Legal, Environmental.

LOS 44.e: Competitive Strategies:

- Cost leadership: Lowest costs, lowest prices, high volume
- Differentiation: Distinctive products, price premium
- Focus: Target niche market

19 Company Analysis: Forecasting

LOS 45.a: Forecast Approaches:

1. Historical results (stable, noncyclical)
2. Base rate convergence (converge to industry average)
3. Management guidance (when proven track record)
4. Discretionary (cyclical, transitional, unique)

LOS 45.b: Revenue Forecasting:

- Top-down: GDP growth or market growth × market share
- Bottom-up: Price × Volume, segment revenues, capacity-based

LOS 45.c: Forecasting Formulas:

$$\boxed{\text{Forecast COGS} = (\text{Historical COGS/Rev}) \times \text{Forecast Rev}}$$

$$\boxed{\text{Forecast AR} = \text{DSO} \times (\text{Forecast Rev}/365)}$$

$$\boxed{\text{Forecast Inventory} = \text{DOH} \times (\text{Forecast COGS}/365)}$$

$$\boxed{\text{Forecast AP} = \text{DPO} \times (\text{Forecast COGS}/365)}$$

LOS 45.e: Scenario Analysis: Use multiple alternative assumptions to examine sensitivity to changes in economic environment, competition, technology.

20 Equity Valuation: Concepts and Basic Tools

20.1 Dividend Discount Models

LOS 46.e: DDM General Form:

$$V_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+k)^t}$$

LOS 46.g: Preferred Stock Valuation:

$$V_0 = \frac{D_p}{k_p}$$

LOS 46.h: Gordon Growth Model:

$$V_0 = \frac{D_1}{k-g}$$

$$g = \text{ROE} \times (1 - \text{Payout Ratio})$$

Example: Gordon growth: D0 = \$1.50, g = 8%, k = 12%. D1 = \$1.62. V0 = \$1.62/(0.12-0.08) = \$40.50.

Two-Stage DDM:

$$V_0 = \sum_{t=1}^{n} \frac{D_0(1+g^*)^t}{(1+k)^t} + \frac{D_{n+1}}{(k-g_c)(1+k)^n}$$

Example: Two-stage: D0 = \$1.00, g* = 15% for 2 years, g = 5%, k = 11%. D1 = \$1.15, D2 = \$1.32, D3 = \$1.39. P2 = \$1.39/(0.11-0.05) = \$23.17. V0 = \$1.15/1.11 + \$24.49/1.2321 = \$20.92.

20.2 Relative Valuation

LOS 46.k: Price Multiples:

$$P/E = \text{Price}_{\overline{\text{EPS}}} \quad P/S = \frac{\text{Price}}{\text{Sales/Share}}$$

$$P/B = \text{Price}_{\overline{\text{BV/Share}}} \quad P/CF = \frac{\text{Price}}{\text{CF/Share}}$$

Justified P/E:

$$P_0 \overline{E_1 = \frac{\text{Payout Ratio}}{k-g}}$$

LOS 46.l: Enterprise Value:

$$\text{EV} = \text{Market Cap} + \text{Market Debt - Cash}$$

$$\text{EV/EBITDA} = \text{EV}_{\overline{\text{EBITDA}}}$$

Example: EV/EBITDA: Market cap \$8M, debt \$1.8M, cash \$0.25M → EV = \$9.55M. EBITDA \$1M → EV/EBITDA = 9.55.

LOS 46.m: Asset-Based Valuation:

$$\text{Equity Value} = \text{Fair Value of Assets - Fair Value of Liabilities}$$

Best for: tangible short-term assets, assets with ready market values, or liquidation.

20.3 Valuation Model

DDM: Advantages: present value theory, widely accepted. Disadvantages: input sensitivity, requires k¿g, firm must pay dividends.

Price Multiples: Advantages: useful for returns prediction, widely used, available. Disadvantages: lagging, comparability issues, meaningless if denominator negative.

Asset-Based: Advantages: floor values, reliable for tangible assets. Disadvantages: market values difficult to obtain, inaccurate for intangible-rich firms.

Formula Summary

Formula	**Description**
$V_0 = \frac{D_1}{k-g}$	Gordon growth model
Basic EPS $= \frac{\text{NI}-\text{Pfd Div}}{\text{WA Shares}}$	Basic EPS
Diluted EPS $= \frac{\text{NI}-\text{Pfd Div}+\text{Convertible Adj}}{\text{WA Shares}+\text{Conversion}}$	Diluted EPS
$\text{FCFF} = \text{CFO} + [\text{Int}(1-t)] - \text{FC}_{\text{INV}}$	Free cash flow to firm
$\text{FCFE} = \text{CFO} - \text{FC}_{\text{INV}} + \text{Net Borrowing}$	Free cash flow to equity
Current Ratio $= \frac{\text{CA}}{\text{CL}}$	Current ratio
Quick Ratio $= \frac{\text{Cash}+\text{MS}+\text{AR}}{\text{CL}}$	Quick ratio
Inventory Turnover $= \frac{\text{COGS}}{\text{Avg Inv}}$	Inventory turnover
Receivables Turnover $= \frac{\text{Revenue}}{\text{Avg AR}}$	Receivables turnover
Debt-to-Equity $= \frac{\text{Total Debt}}{\text{Total Equity}}$	Debt-to-equity
$\text{ROE} = \text{NPM} \times \text{TAT} \times \text{FL}$	DuPont ROE
$\text{EV} = \text{Market Cap} + \text{Market Debt} - \text{Cash}$	Enterprise value
Justified P/E $= \frac{\text{Payout Ratio}}{k-g}$	Justified P/E
Margin Call Price $= P_0 \times \frac{1-IM}{1-MM}$	Margin call price
$g = \text{ROE} \times (1 - \text{Payout Ratio})$	Sustainable growth
Cash Conversion Cycle $= \text{DOH} + \text{DSO} - \text{DPO}$	CCC

Annexure: Numerical Graphs

21 Financial Statement Analysis Visualizations

21.1 FIFO vs LIFO Analysis During Inflation

21.2 DuPont Analysis - ROE Decomposition

21.3 Gordon Growth Model Sensitivity Analysis

21.4 Capitalizing vs Expensing Impact

21.5 Financial Ratios Analysis & Benchmarking

21.6 Valuation Multiples Across Sectors

21.7 Financial Statement Analysis Framework

21.8 Porter's Five Forces Framework

21.9 Earnings Quality Spectrum

21.10 Inventory Turnover Analysis

21.11 Financial Leverage and Solvency Analysis

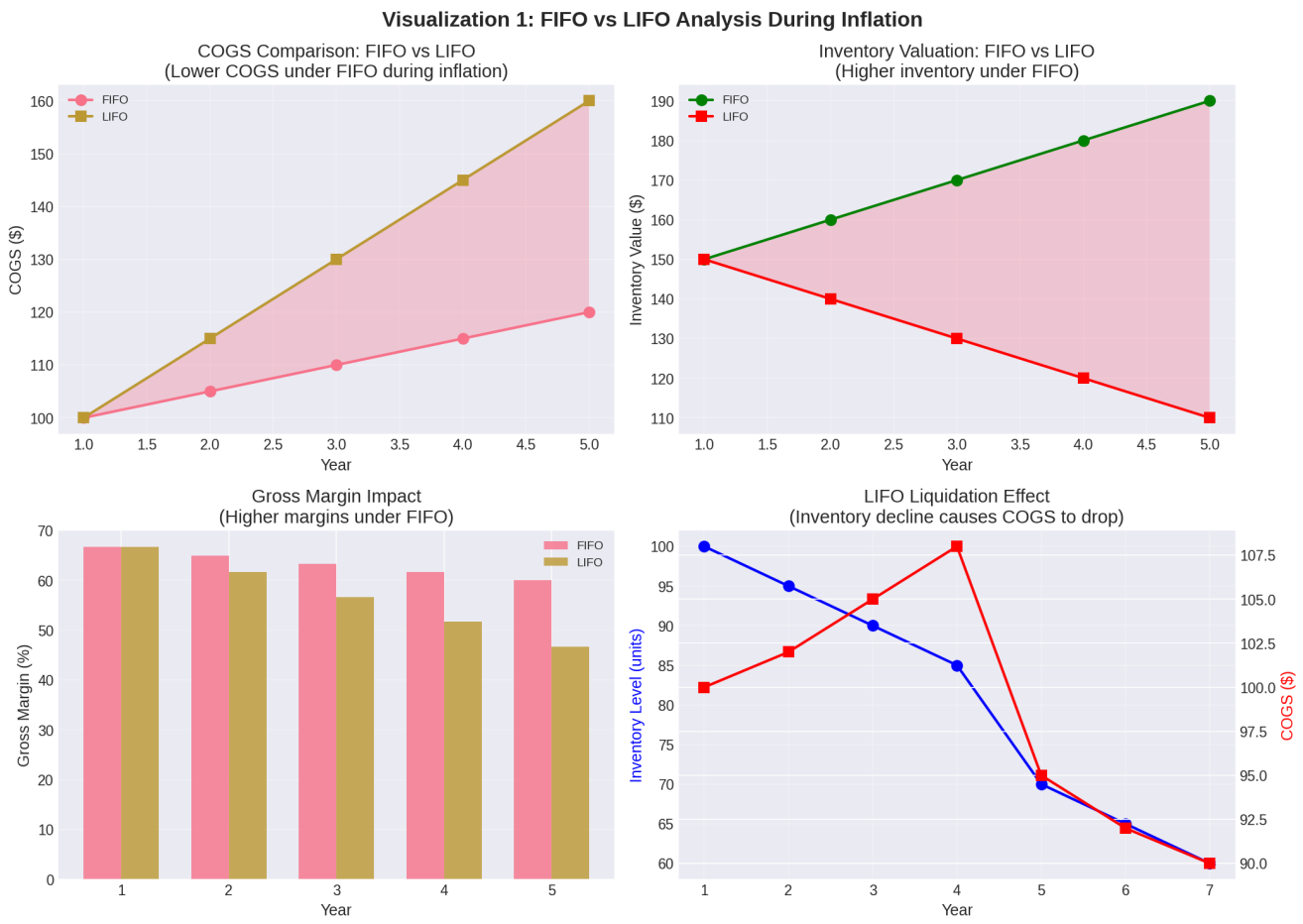

Figure 1: FIFO vs LIFO Analysis: COGS, Inventory, Gross Margin, and LIFO Liquidation Effect

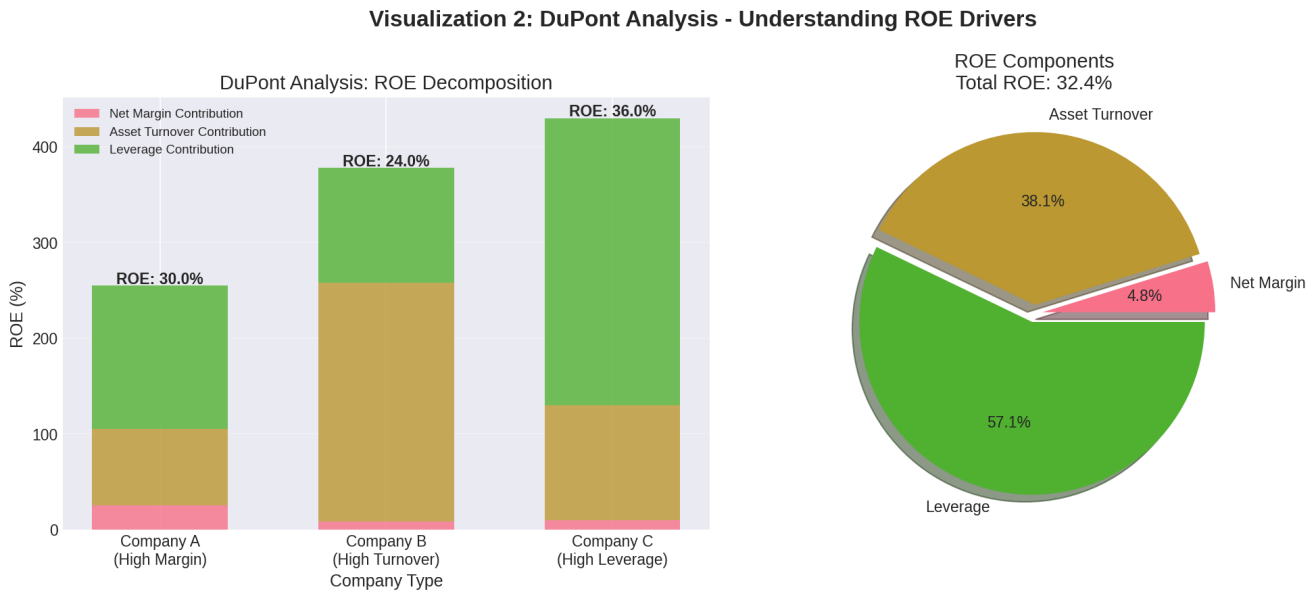

Figure 2: DuPont Analysis: Understanding ROE Drivers (3-Way Decomposition)

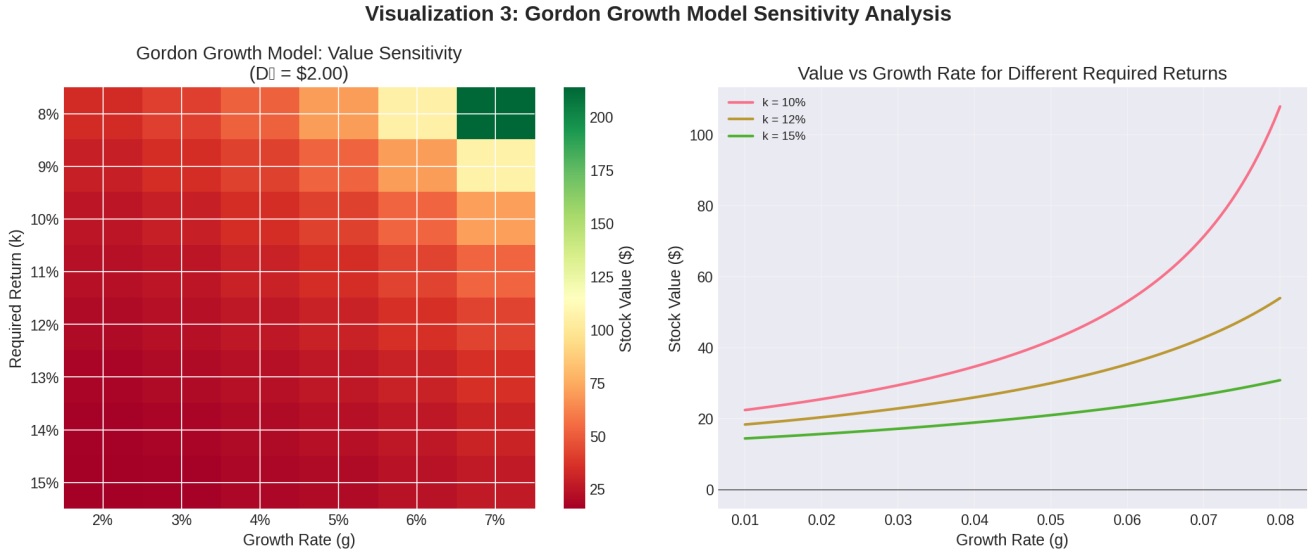

Figure 3: Gordon Growth Model: Value Sensitivity to Growth Rate and Required Return

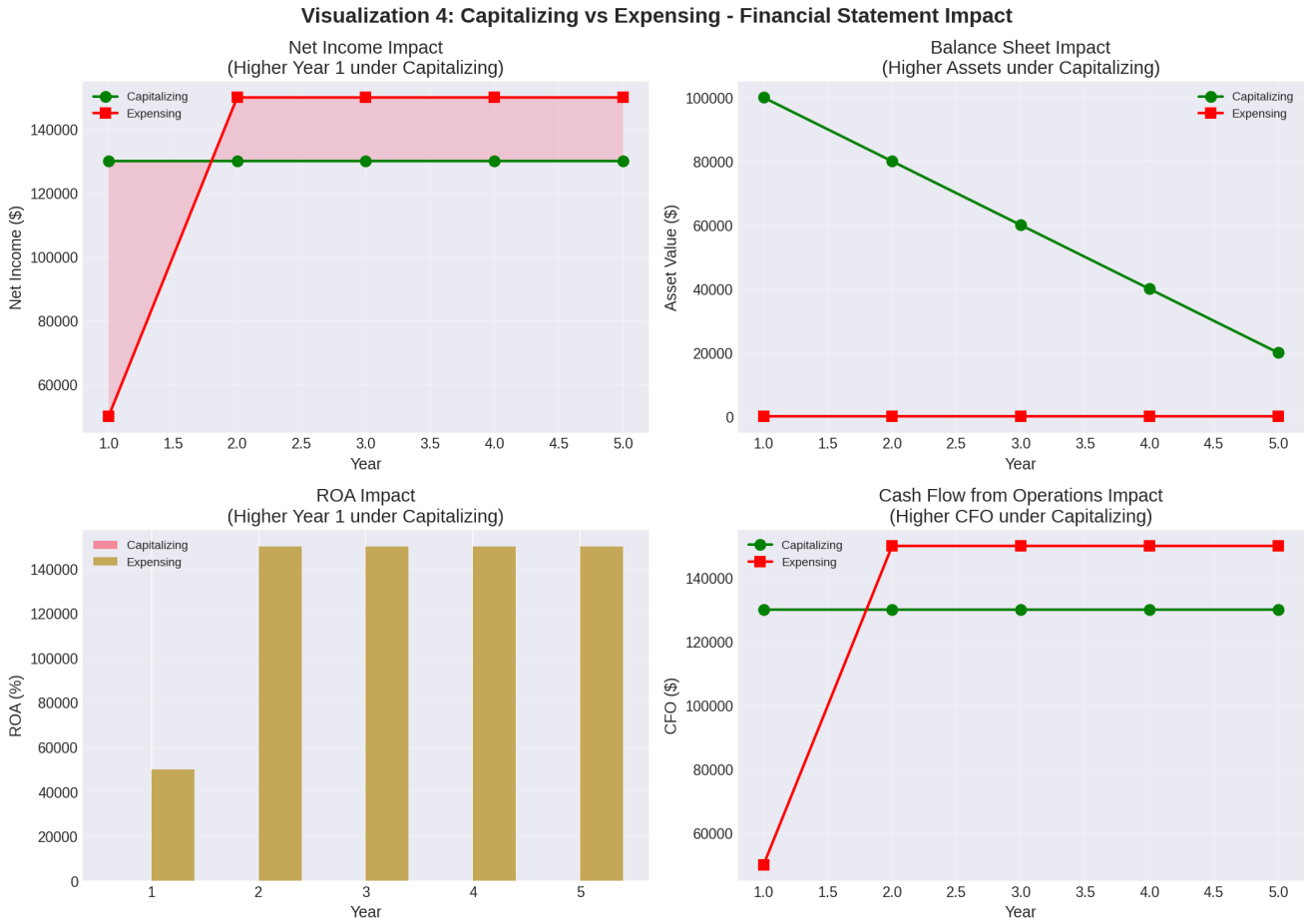

Figure 4: Capitalizing vs Expensing: Financial Statement Impact (NI, Assets, ROA, CFO)

Figure 5: Financial Ratios Analysis: Industry Benchmarking, Cash Conversion Cycle, P/E Distributions

Figure 6: Valuation Multiples: P/E by Sector, EV/EBITDA by Industry, P/B vs ROE, PEG Ratios

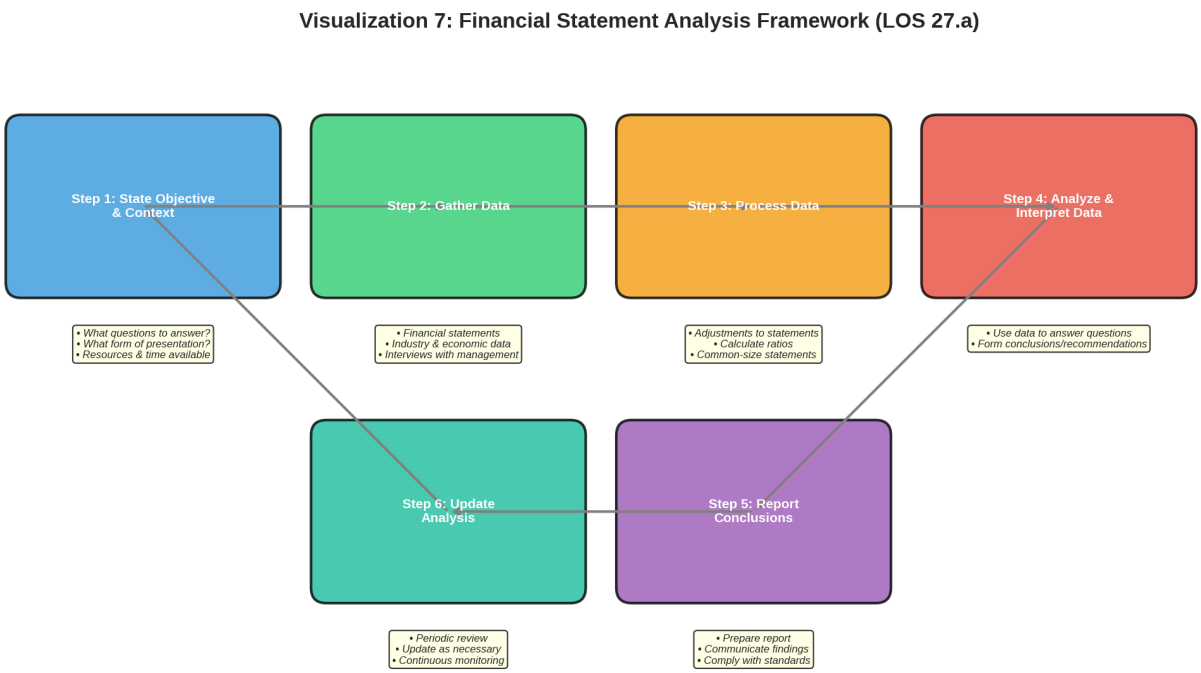

Figure 7: 6-Step Financial Statement Analysis Framework (LOS 27.a)

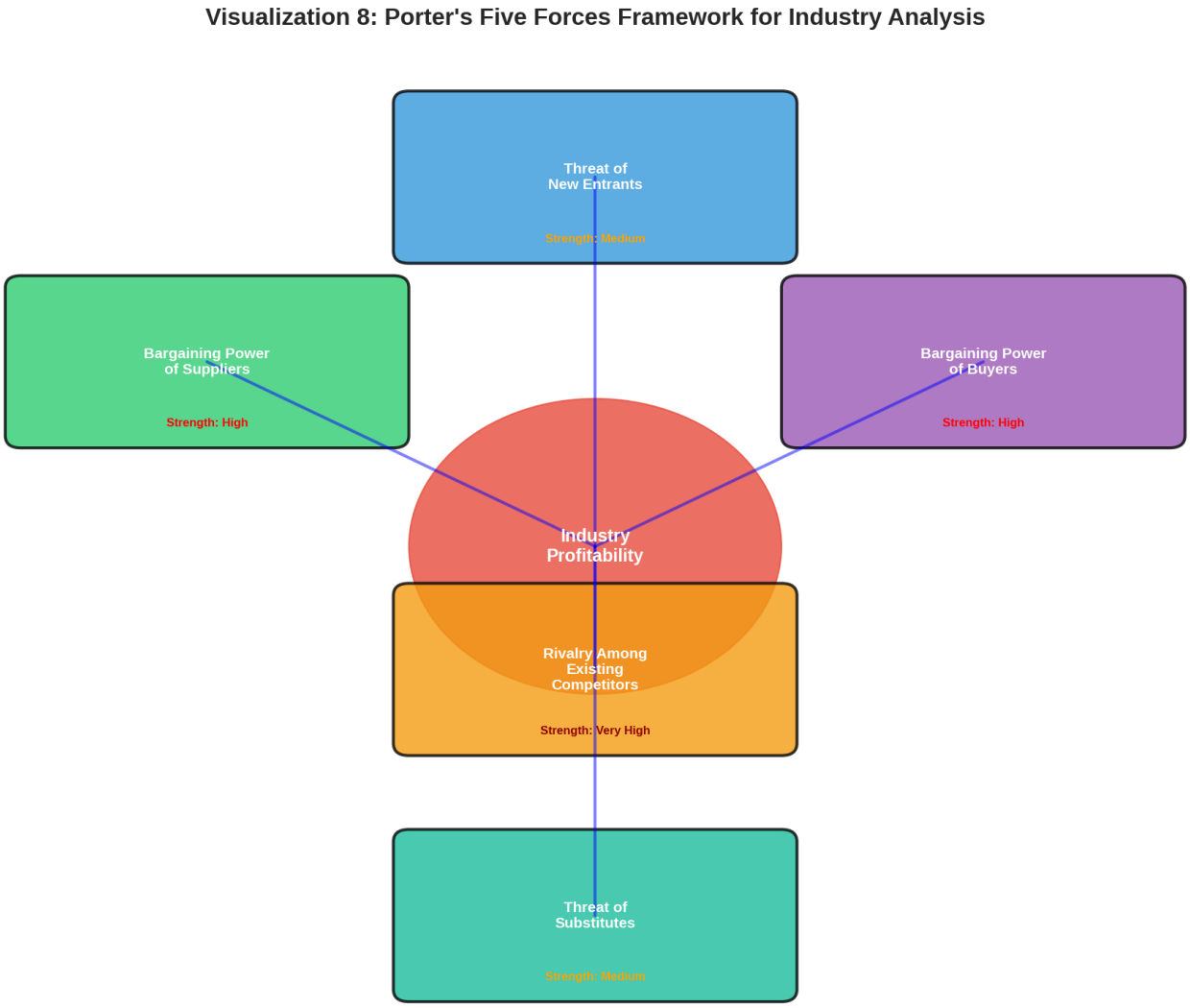

Figure 8: Porter's Five Forces Framework for Industry Analysis

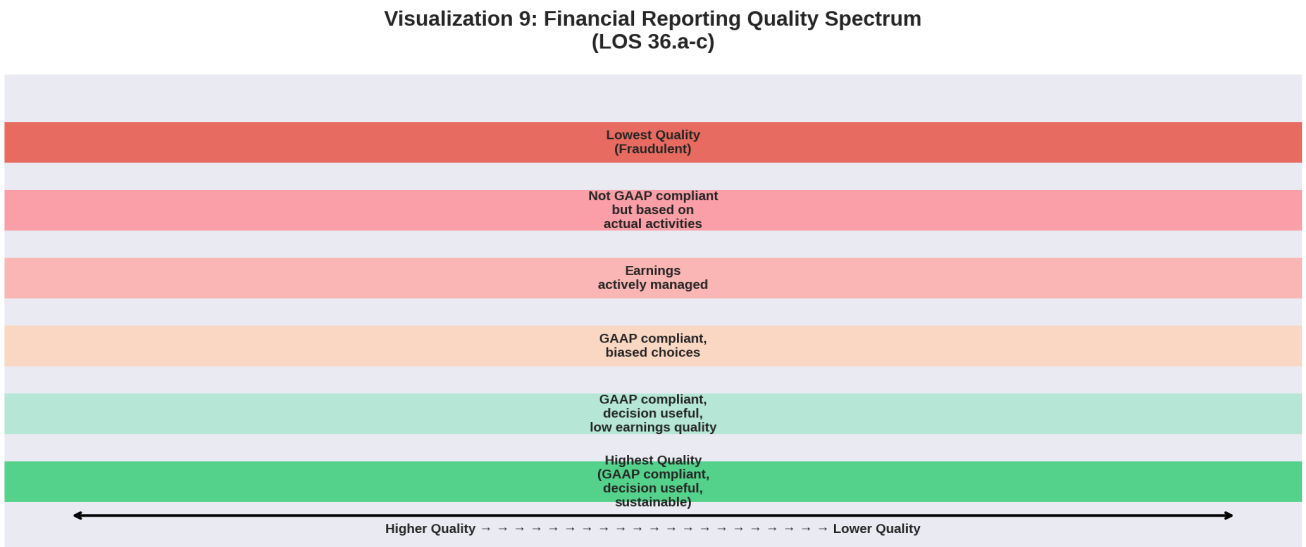

Figure 9: Financial Reporting Quality Spectrum (LOS 36.a-c)

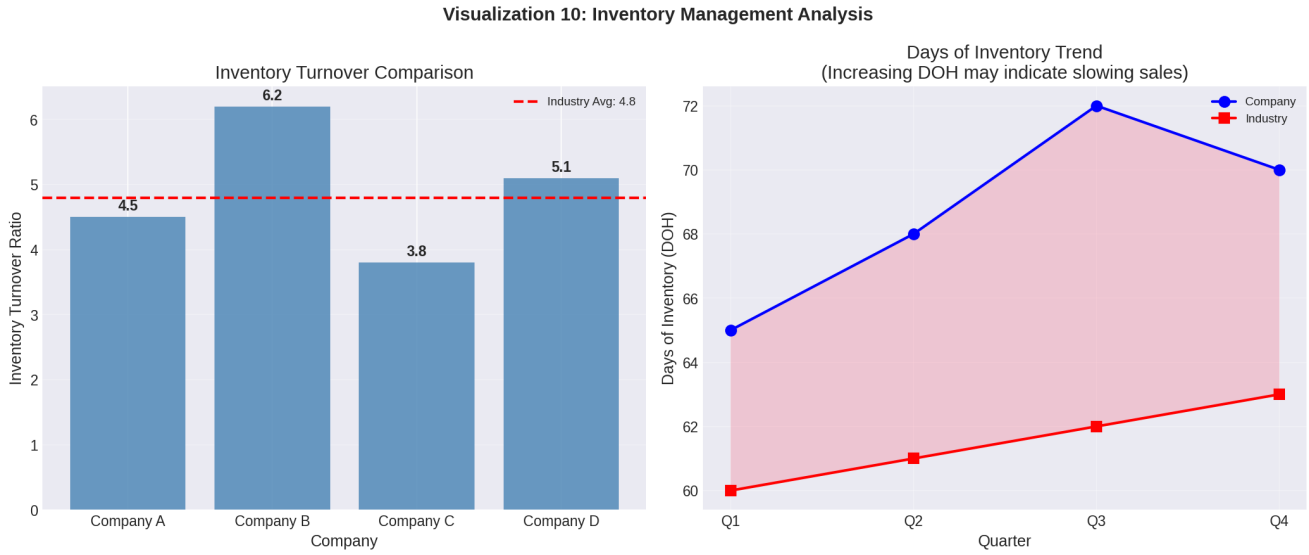

Figure 10: Inventory Management Analysis: Turnover Comparison and Days of Inventory Trend

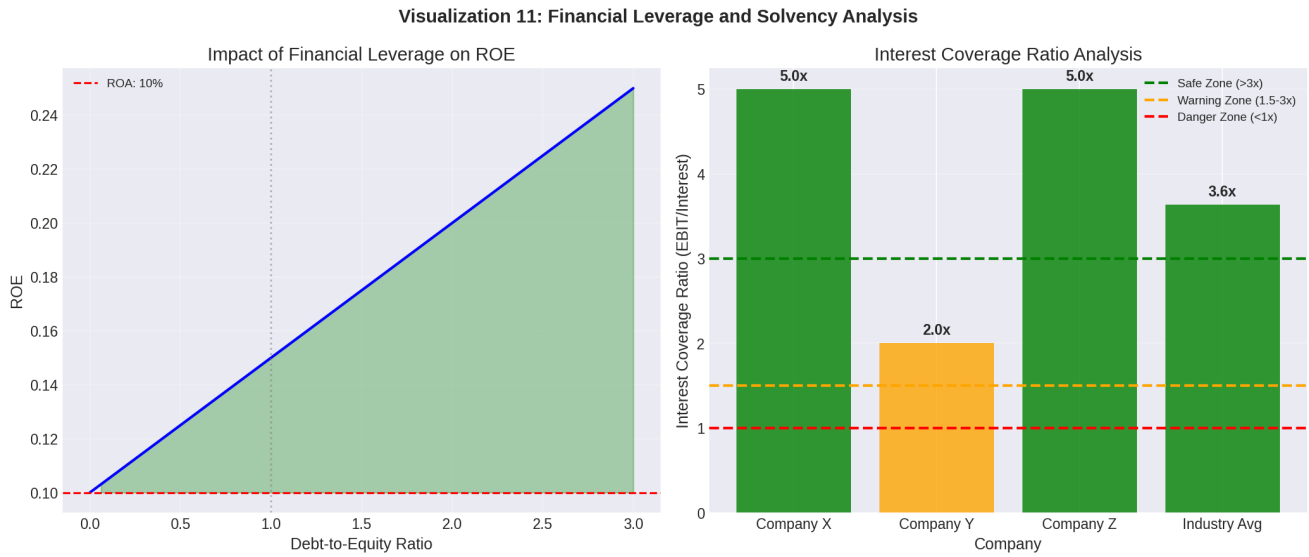

Figure 11: Financial Leverage Analysis: ROE Impact and Interest Coverage Ratios